



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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CSA-0211 C-PROGRAMMING

Emergency Ambulance Dispatch Management System Using C

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ABSTRACT

- Emergency Ambulance Dispatch Management System is developed using C programming to manage emergency ambulance services efficiently.
- The system performs patient registration by storing details such as name, age, address, contact number, and emergency type.
- It allocates the nearest available ambulance to the patient, reducing response time during emergencies.
- The system uses priority-based emergency processing, ensuring that critical patients are served before less urgent cases.
- This project improves emergency response, ambulance utilisation patient care, and hospital efficiency, helping to save valuable time and lives.



PROBLEM STATEMENT

- Many hospitals still use manual methods to register patients and dispatch ambulances, causing delays during emergencies. There is no efficient system to quickly identify and allocate the nearest available ambulance. When multiple emergency requests arrive at the same time, critical patients may not receive immediate attention due to the lack of priority-based processing.
- Manual ambulance management can lead to longer response times, poor resource utilisation, and increased workload for hospital staff.



LITERATURE SURVEY



Paper Title	Key Findings	Limitation
Patient Registration System	Registers patient details and maintains patient records efficiently.	Does not support ambulance allocation or emergency prioritization.
Ambulance Dispatch Management System	Allocates available ambulances to patients and reduces response time.	Does not include patient registration or priority-based emergency processing.
Emergency Priority Management System	Classifies emergency cases based on severity and gives priority to critical patients.	Does not integrate patient registration and ambulance allocation into a single system.

Research Gap

- Existing systems focus only on individual emergency service functions.
- Limited integration of patient registration, ambulance allocation, and priority-based emergency processing.
- Our project combines all these features into one application using **C programming, structures, functions, and file handling.**



OBJECTIVE

- **Patient Registration**

To register patient details such as name, age, address, contact number, and emergency type accurately and quickly, ensuring proper record management and faster emergency response.

- **Ambulance Allocation**

To identify available ambulances and allocate the nearest ambulance to the patient, reducing response time and improving ambulance utilization during emergencies.

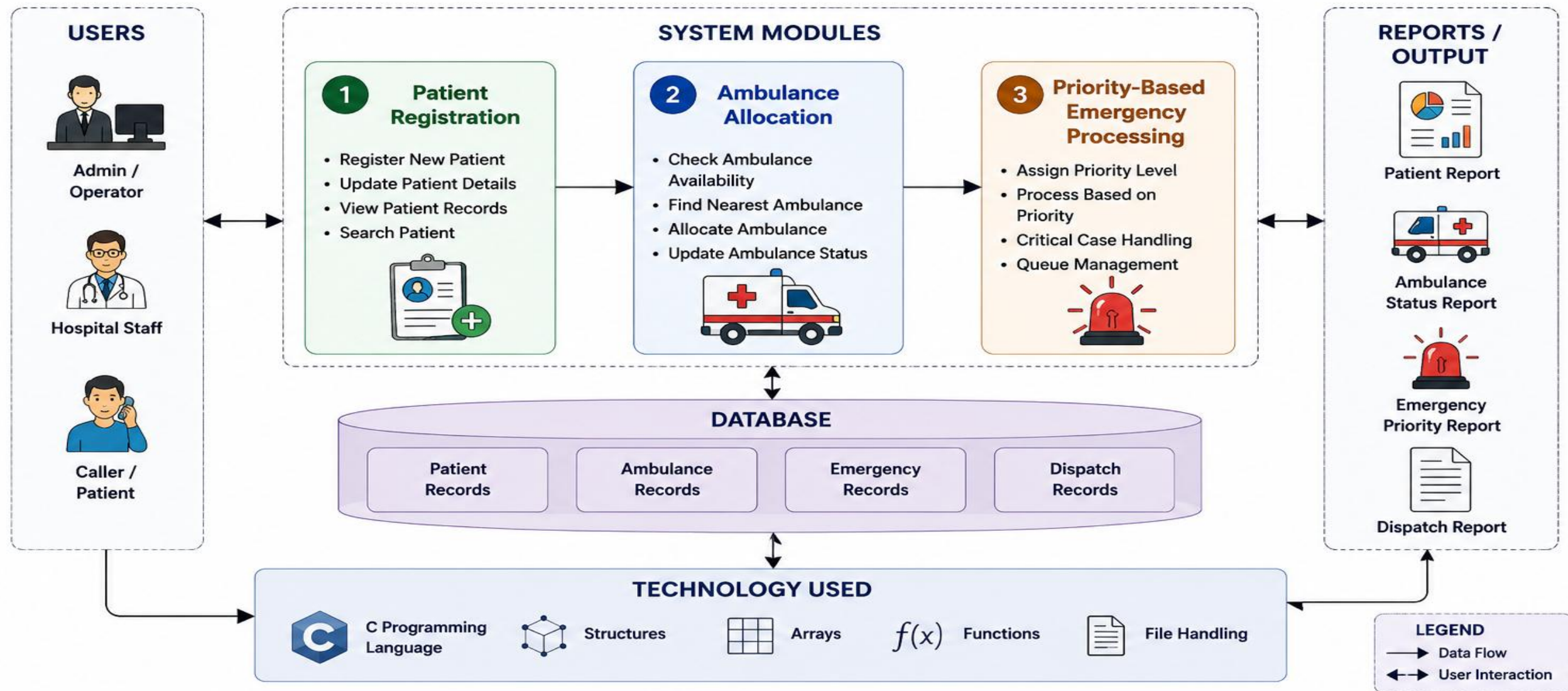
- **Priority-Based Emergency Processing**

To process emergency requests based on the severity of the patient's condition, ensuring that critical cases receive immediate medical attention and improving the efficiency of emergency healthcare services..



SYSTEM ARCHITECTURE

Emergency Ambulance Dispatch Management System with Patient Registration, Ambulance Allocation, and Priority-Based Emergency Processing Using C





MODULE -1 Patient Registration Module



1. Register New Patient – Enter patient ID, name, age, contact number, and emergency details.
2. Store Patient Details – Saves patient information in the patient records.
3. Update Patient Information – Allows staff to modify existing patient details.
4. Search Patient Records – Quickly finds patient information using the patient ID or name.

The screenshot displays a 'NEW PATIENT' registration form with a dark blue background. The form includes the following fields and options:

- NAME:** A text input field with the placeholder 'Full name'.
- AGE:** A text input field with the placeholder 'Age'.
- CONTACT:** A text input field with the placeholder 'Phone number'.
- LOCATION:** A text input field with the placeholder 'Current address'.
- CONDITION:** A text input field with the placeholder 'Emergency description'.
- PRIORITY:** Three radio button options: 'CRITICAL', 'SERIOUS' (which is selected and highlighted with a yellow border), and 'MINOR'.
- Register Patient:** A large blue button at the bottom of the form.

Output-1

```
p.patientID = nextPatientID++;

printf("Enter Patient Name: ");
fgets(p.name, sizeof(p.name), stdin);
p.name[strcspn(p.name, "\n")] = '\0';

printf("Enter Age: ");
scanf("%d", &p.age);
clearBuffer();

printf("Enter Contact Number: ");
fgets(p.contact, sizeof(p.contact), stdin);
p.contact[strcspn(p.contact, "\n")] = '\0';

printf("Enter Location: ");
fgets(p.location, sizeof(p.location), stdin);
p.location[strcspn(p.location, "\n")] = '\0';

printf("Enter Emergency Condition: ");
fgets(p.condition, sizeof(p.condition), stdin);
p.condition[strcspn(p.condition, "\n")] = '\0';

printf("\nSelect Priority:\n");
printf("1. CRITICAL\n");
printf("2. SERIOUS\n");
printf("3. MINOR\n");
printf("Enter Priority: ");
scanf("%d", &p.priority);
clearBuffer();

if (p.priority < 1 || p.priority > 3)
{
    printf("Invalid priority! Setting to SERIOUS.\n");
    p.priority = 2;
}
```

```
=====
EMERGENCY AMBULANCE SYSTEM
=====
Click to collapse the range.
=====
1. Register Patient
2. Display Patients
3. Exit

Enter your choice: 1

=====
PATIENT REGISTRATION
=====
Enter Patient Name: kumar
Enter Age: 25
Enter Contact Number: 99874561425
Enter Location: annanager,chennai
Enter Emergency Condition: critical

Select Priority:
1. CRITICAL
2. SERIOUS
3. MINOR
Enter Priority: 2

=====
PATIENT REGISTERED SUCCESSFULLY
=====
Patient ID : 1001
Name       : kumar
Age        : 25
Contact    : 99874561425
Location   : annanager,chennai
Condition  : critical
Priority    : SERIOUS
Status     : WAITING
=====
```




Module – 2 Ambulance Allocation



1. Check Ambulance Availability – Identifies ambulances that are currently available for dispatch.
2. Find Nearest Ambulance – Selects the nearest available ambulance for the patient.
3. Allocate Ambulance – Assigns the selected ambulance to the emergency case.
4. Update Ambulance Status – Changes the ambulance status to Dispatched/Busy after allocation.

NEW AMBULANCE

VEHICLE NUMBER
e.g. TN-09-AB-1234

DRIVER NAME
Full name

DRIVER CONTACT
Phone number

Add to Fleet

Output-2

```
=====
      AMBULANCE MANAGEMENT MODULE
=====
1. Add Ambulance Details
2. Show Available Ambulances
3. Find Ambulances Near Hospital
4. Display All Ambulances
5. Exit

Enter your choice: 1

=====
      ADD AMBULANCE DETAILS
=====
Enter Vehicle Number: #178
Enter Driver Name: ravi
Enter Driver Contact: 963258741
Enter Current Location: chittepeddu
Enter Nearby Hospital: saveetha

Ambulance added successfully!
Ambulance ID : 1
```

```
Enter your choice: 2

=====
      AVAILABLE AMBULANCES
=====

Ambulance ID : 1
Vehicle No   : #178
Driver Name  : ravi
Contact      : 963258741
Location     : chittepeddu
Hospital     : saveetha
Status       : AVAILABLE

-----
```

```
Enter your choice: 3

=====
      AMBULANCES NEAR HOSPITAL
=====
Enter Hospital Name: saveetha

Ambulance ID : 1
Vehicle No   : #178
Driver Name  : ravi
Contact      : 963258741
Location     : chittepeddu
Hospital     : saveetha
Status       : AVAILABLE

-----
```

```
Enter your choice: 4

=====
      ALL AMBULANCES
=====

Ambulance ID : 1
Vehicle No   : #178
Driver       : ravi
Contact      : 963258741
Location     : chittepeddu
Hospital     : saveetha
Status       : AVAILABLE

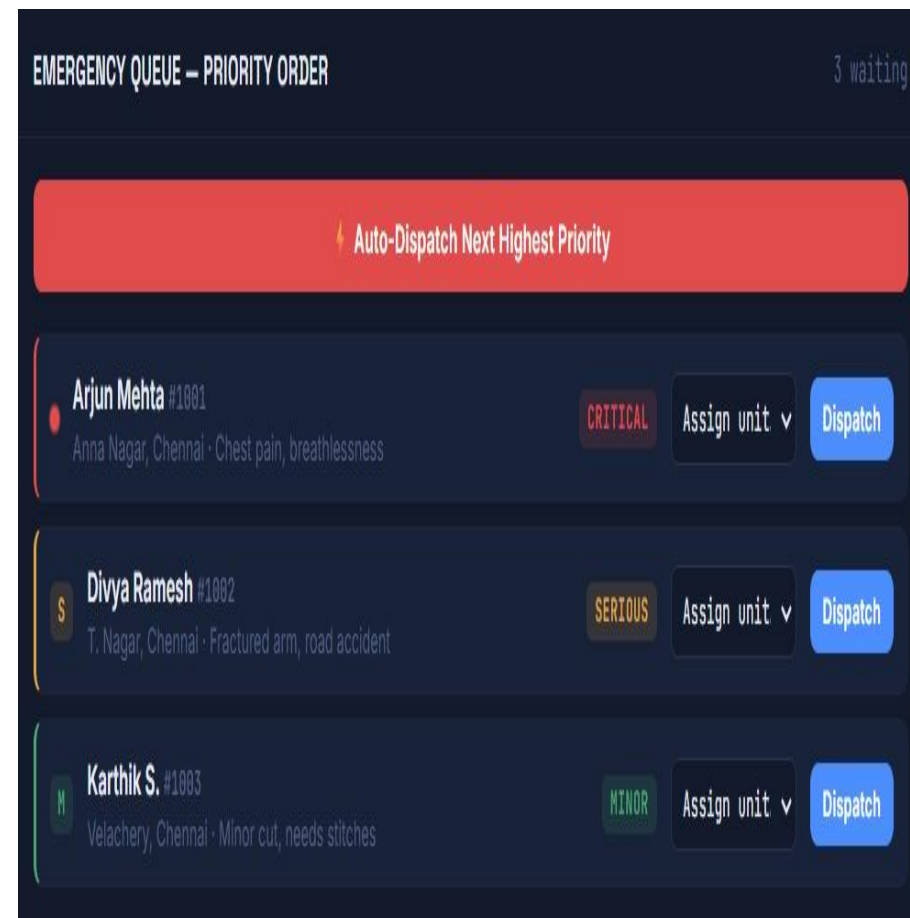
-----
```



Module -3 Priority-Based Emergency Processing



- Assign Priority Level – Classifies patients as Critical, Serious, or Normal based on emergency condition.
- Process by Priority – Handles critical cases first to provide faster emergency assistance.
- Manage Emergency Queue – Maintains and processes patients in the appropriate priority order.
- Dispatch Ambulance – Sends the allocated ambulance to high-priority emergency cases and updates the dispatch status.





CONCLUSION



- Efficient Patient Management – The system provides quick registration, updating, and searching of patient records.
- Fast Ambulance Allocation – It identifies available and nearest ambulances and assigns them efficiently.
- Priority-Based Processing – Critical emergencies are handled first based on their priority level.
- Improved Emergency Response – The system helps reduce delays and improves coordination between patients, hospital staff, and ambulance services.
- Reliable Record Management – Patient, ambulance, emergency, and dispatch records are maintained systematically for easy tracking and reporting.

The image features a central text 'thank you' in a black, cursive script font. This text is surrounded by a decorative border of autumn-themed elements. The border includes several large, detailed leaves in shades of red, orange, and yellow. Interspersed among the leaves are clusters of small, dark blue berries and a few brown, textured nuts. The entire composition is set against a plain, light blue background.

thank you